Md Nahidul Islam

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personal website | Google Scholar | ORCID | Research Gate | GitHub | LinkedIn

Research Interests

Autonomous driving, interactive motion planning, trajectory forecasting, trustworthy machine learning, interaction representation evaluation, counterfactual risk certification, model compression, knowledge distillation, and safe robotics.

Research Profile: Ph.D. researcher developing scalable methods to evaluate learned representations, certify decision-making risk, and improve the efficiency of safety-critical autonomous-driving systems.

Education

Doctor of Philosophy (Ph.D.) in Computer Science *2024 – Present*
UNIVERSITY OF MEMPHIS Memphis, TN, USA
Advisor: Dr. Myounggyu Won | GPA: 3.95/4.00 | M.S. earned en route, May 2026
M.S. Thesis: BucketKD: A Safety-Aware Bucket-Based Knowledge Distillation Framework for End-to-End Motion Planning

Bachelor of Science (B.Sc.) in Computer Science and Engineering *2012 – 2017*
BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY (BUET) Dhaka, Bangladesh

Publications and Manuscripts

Full list available at: Google Scholar | ORCID | Research Gate

*Author of this CV in **bold**. Submitted manuscripts are clearly marked as under review.*

[1] L. Won, N. H. Nguyen, **M. N. Islam**, L. Das, and M. Won, "LiDARBridge: Range-Dependent LiDAR Noise Modeling and Prototype Alignment for Simulation-to-Real 3D Object Detection," *submitted to the Asian Conference on Computer Vision (ACCV) 2026.*

[2] **M. N. Islam**, M. Won, "What Aggregate Metrics Miss: A Residual-Signal Protocol for Interaction Representations in Trajectory Forecasting," *submitted to the Conference on Robot Learning (CoRL) 2026.*

[3] **M. N. Islam**, M. Won, "CRiC: Protocol Scoped Counterfactual Risk Certification for Interactive Driving," *submitted to NeurIPS 2026.*

[4] **M. N. Islam**, M. H. Ali, D. Dasgupta, and M. Won, "BucketKD: A Safety-Aware Bucket-Based Knowledge Distillation Framework for End-to-End Motion Planning," *submitted to AAAI 2027.*

[5] **M. N. Islam**, "BucketKD: A Safety-Aware Bucket-Based Knowledge Distillation Framework for End-to-End Motion Planning," M.S. thesis, Dept. of Computer Science, University of Memphis, Memphis, TN, USA, 2026. [link]

[6] L. Hwang, **M. N. Islam**, M. H. Ali, D. Dasgupta, and M. Won, "Detecting Stealthy Anomalies with Autoencoders Using Windowed Variance-Aware Loss Functions," in *Proc. IEEE Military Communications Conference (MILCOM)*, 2025. [DOI]

[7] N. Sadeq, S. Ahmed, S. S. Shubha, **M. N. Islam**, and M. A. Adnan, "Bangla Voice Command Recognition in End-to-End System Using Topic Modeling Based Contextual Rescoring," in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2020.

[8] S. Ahmed, N. Sadeq, S. S. Shubha, **M. N. Islam**, and M. A. Adnan, "Preparation of Bangla Speech Corpus from Publicly Available Audio & Text," in *Proc. 12th Int. Conf. Language Resources and Evaluation (LREC)*, 2020, pp. 6586–6592.

[9] S. S. Shubha, N. Sadeq, S. Ahmed, **M. N. Islam**, M. A. Adnan, M. Y. A. Khan, and M. Z. Islam, "Customizing Grapheme-to-Phoneme System for Non-Trivial Transcription Problems in Bangla Language," in *Proc. NAACL-HLT*, 2019, pp. 3192–3201.

Selected Projects

Residual-Signal Protocol (RSP) for Interaction Representations *2026 – Present*
PH.D. RESEARCH PROJECT | PYTORCH, ARGOVERSE 2, MTR, QCNET, REPRESENTATION LEARNING

- Developed an evaluation framework to test whether learned target–neighbor interaction representations explain trajectory-forecasting errors beyond physical interaction metrics and scenario covariates.
- Designed **DyadCode**, a discrete representation of pairwise driving interactions, and evaluated its residual explanatory signal against random, shuffled, and K-means-based controls using robustness tests and PET/TTC ablations.
- Evaluated the framework on Argoverse 2 using MTR and QCNet, including analyses of representation quality, residual consistency, ranking differences, and behavior across interaction conditions.

CRiC: Counterfactual Risk Certification for Interactive Driving *2026 – Present*
PH.D. RESEARCH PROJECT | PYTORCH, TRAJECTORY FORECASTING/PLANNING, CONFORMAL CALIBRATION

- Developed a protocol-scoped certification framework for interactive autonomous driving that evaluates a frozen decision protocol rather than only a predictor.
- Designed calibrated upper-risk-bound selection and abstention logic for safety-aware candidate choice under uncertainty.
- Built streaming-friendly evaluation artifacts and reliability analyses for large-scale scene/candidate rollouts.

BucketKD: Safety-Aware End-to-End Motion Planner Compression *Jan. 2024 – Feb. 2026*
M.S. Thesis, University of Memphis | PyTorch, CARLA Simulator

- Developed a bucket-based knowledge-distillation framework for compressing end-to-end autonomous-driving planners while preserving safety-relevant planning behavior.
- Designed TTC-based waypoint attention and adaptive bucket construction to improve safety-aware distillation.

Research Experience

Graduate Research Assistant *Jan. 2024 – Present*
University of Memphis – Advisor: Prof. Myounggyu Won — Memphis, TN, USA

- Developing risk-aware evaluation and decision-making methods for interactive autonomous driving.
- Designed TTC-guided attention and adaptive state discretization to preserve safety-critical behavior during motion-planner distillation.
- Developed methods to evaluate whether learned interaction representations explain trajectory-forecasting errors beyond physical and scenario-based factors.
- Built scalable autonomous-driving research pipelines using PyTorch, CARLA, WOMD, Argoverse 2, and HPC/GPU infrastructure.

Research Assistant *Aug. 2018 – Jan. 2020*
Bangladesh University of Engineering and Technology (BUET) — Dhaka, Bangladesh

- Conducted research on Bangla speech recognition, low-resource language modeling, and grapheme-to-phoneme conversion.
- Contributed to peer-reviewed publications at ICASSP, LREC, and NAACL.
- Built and evaluated sequence-to-sequence and attention-based models for Bangla ASR and text/speech processing .

Industry Experience

Assistant Director, Enterprise Data Warehouse *Nov. 2020 – Dec. 2023*
Bangladesh Bank (Central Bank of Bangladesh) — Dhaka, Bangladesh

- Designed and maintained ETL pipelines and internal analytics platforms for enterprise-scale regulatory financial data.
- Developed web applications and proof-of-concept ML tools using Django, React, Oracle, Docker, and CI/CD workflows.

Software Engineer, AI Solutions Team *Feb. 2020 – Oct. 2020*
Binate Solutions Ltd. — Dhaka, Bangladesh

- Developed REST APIs, ML-backed services, and SaaS web applications using Django, TensorFlow, and cloud deployment tools.

Information Analyst, MIS Department *Feb. 2017 – Mar. 2018*
ACI Ltd. — Dhaka, Bangladesh

- Built reporting dashboards and applied machine-learning methods to business and operations datasets.

Teaching, Mentoring, and Outreach

Graduate Teaching Assistant *Jan. 2024 – Present*
University of Memphis, Department of Computer Science — Memphis, TN, USA

- Mentored students in *Capstone Software Project* and *Operating Systems* courses with enrollments of 50–80 students; evaluated projects and proctored exams.

Technical Mentor and Volunteer *2024–Present*
Autonomous R/C Cars Cybersecurity Summer Camp — Memphis, TN, USA

- Serve as a recurring technical mentor for an annual K–12 summer camp focused on introductory Python programming, autonomous driving, and automotive cybersecurity.
- Help students collect driving data, train models, and deploy autonomous-driving systems on Raspberry Pi-based Donkey Car platforms.
- Support supervised demonstrations of password cracking, buffer-overflow vulnerabilities, and man-in-the-middle attacks to explain cybersecurity risks in connected autonomous vehicles.

Technical Skills

Programming / Web Python, C++, C, C#, Java, JavaScript, SQL/PL-SQL, R; Django, FastAPI, React, Redux, Node.js, ASP.NET
AI / Robotics PyTorch, TensorFlow, Hugging Face, OpenCV; CARLA, Donkey Car, Arduino Uno, Raspberry Pi, trajectory

forecasting, motion planning, TTC safety metrics, WOMD, AV2

Systems / Data Linux, Git, Docker, HPC/GPU clusters, AWS, DigitalOcean, CI/CD; Oracle, PostgreSQL, MySQL, MS SQL Server

Awards and Honors

- **1st Place, Individual Category**, Tech901 CTF 2025, Online, Oct. 2025.
- **1st Position, Team Category**, Financial Institutions Cyber Drill 2021, organized by BGD e-GOV CIRT, ICT Division, Bangladesh.
- **Letter of Appreciation**, awarded by the Governor of Bangladesh Bank, 2021.
- **Technical Scholarship**, Bangladesh University of Engineering and Technology, 2012.